Battle of the Sexes

***p. 182* … how much more severe must be the conflict between mates, who are not related to each other?**

As so often, this opening sentence hides an implicit 'other things being equal'. Obviously mates are likely to have a great deal to gain from cooperation. This emerges again and again throughout the chapter. After all, mates are likely to be engaged in a nonzero sum game, a game in which both can increase their winnings by cooperating, rather than one's gain necessarily being the other's loss (I explain this idea in Chapter 12). This is one of the places in the book where my tone swung too far towards the cynical, selfish view of life. At the time it seemed necessary, since the dominant view of animal courtship had swung far in the other direction. Nearly universally, people had uncritically assumed that mates would cooperate unstintingly with each other. The possibility of exploitation wasn't even considered. In this historical context the apparent cynicism of my opening sentence is understandable, but today I would adopt a softer tone. Similarly, at the end of this chapter my remarks about human sexual roles now seem to me naïvely worded. Two books that go more thoroughly into the evolution of human sex differences are Martin Daly and Margo Wilson's *Sex, Evolution, and Behavior*, and Donald Symons's *The Evolution of Human Sexuality*.

***p. 184* … the number of children a male can have is virtually unlimited. Female exploitation begins here.**

It now seems misleading to emphasize the disparity between sperm and egg size as the basis of sex roles. Even if one sperm is small and cheap, it is far from cheap to make millions of sperms and successfully inject them into a female against all the competition. I now prefer the following approach to explaining the fundamental asymmetry between males and females.

Suppose we start with two sexes that have none of the particular attributes of males and females. Call them by the neutral names *A* and *B*. All we need specify is that every mating has to be between an *A* and a *B*. Now, any animal, whether an *A* or a *B*, faces a trade-off. Time and effort devoted to fighting with rivals cannot be spent on rearing existing offspring, and vice versa. Any animal can be expected to balance its effort between these rival claims. The point I am about to come to is that the *A*s may settle at a different balance from the *B*s and that, once they do, there is likely to be an escalating disparity between them.

To see this, suppose that the two sexes, the *A*s and the *B*s, differ from one another, right from the start, in whether they can most influence their success by investing in children or by investing in fighting (I'll use 'fighting' to stand for all kinds of direct competition within one sex). Initially the difference between the sexes can be very slight, since my point will be that there is an inherent tendency for it to grow. Say the *A*s start out with fighting making a greater contribution to their reproductive success than parental behaviour does; the *B*s, on the other hand, start out with parental behaviour contributing slightly more than fighting to variation in *their* reproductive success. This means, for example, that although an *A* of course benefits from parental care, the difference between a successful carer and an unsuccessful carer among the *A*s is smaller than the difference between a successful fighter and an unsuccessful fighter among the *A*s. Among the *B*s, just the reverse is true. So, for a given amount of effort, an *A* can do itself good by fighting, whereas a *B* is more likely to do itself good by shifting its effort away from fighting and towards parental care.

In subsequent generations, therefore, the *A*s will fight a bit more than their parents, the *B*s will fight a bit less and care a bit more than their parents. Now the difference between the best *A* and the worst *A* with respect to fighting will be even greater, the difference between the best *A* and the worst *A* with respect to caring will be even less. Therefore an *A* has even more to gain by putting